

Question 1 - Square Numbers

Create a list from 1 to 20.

Use `map()` to generate squares.

Question 2 - Cube Numbers

Use `map()` to generate cubes of numbers from 1 to 20.

Question 3 - Convert Strings to Uppercase

Given:

```
["python", "java", "go", "rust"]
```

Convert all strings to uppercase using `map()`.

Question 4 - String Length

Find the length of every word using `map()`.

Question 5 - Celsius to Fahrenheit

Convert a list of temperatures from Celsius to Fahrenheit using `map()`.

Question 6 - Filter Even Numbers

Create a list from 1 to 100.

Use `filter()` to keep only even numbers.

Question 7 - Filter Prime Numbers

Create a function to check prime numbers.

Use `filter()` to extract all prime numbers from 1 to 100.

Question 8 - Filter Long Words

Given a list of words, keep only words having more than five characters.

Question 9 - Filter Positive Numbers

Given:

```
[-5, -2, 0, 3, 5, 7, -1, 8]
```

Extract only positive numbers.

Question 10 - List Comprehension Basics

Generate numbers from 1 to 50 using list comprehension.

Question 11 - Even Squares

Generate squares of all even numbers from 1 to 100 using list comprehension.

Question 12 - Multiplication Table

Generate the table of 7 using list comprehension.

Expected Output:

```
7
14
21
...
70
```

Question 13 - Flatten Nested List

Given:

```
[[1,2],[3,4],[5,6]]
```

Convert it into:

```
[1,2,3,4,5,6]
```

using list comprehension.

Question 14 - Dictionary Comprehension

Create a dictionary where:

```
1 → 1  
2 → 4  
3 → 9  
...  
20 → 400
```

Question 15 - Reverse Dictionary

Given:

```
{  
    "A":1,  
    "B":2,  
    "C":3  
}
```

Reverse keys and values using dictionary comprehension.

Question 16 - Set Comprehension

Generate a set containing squares of numbers from 1 to 20.

Question 17 - Combined Practice

Given:

```
numbers = list(range(1,101))
```

Perform the following:

- Filter even numbers
- Square them
- Store the result in a list

Use both `filter()` and `map()`.

Question 18 - Mini Project

Create a Student Result Analyzer.

Input:

- Student names
- Marks

Requirements:

- Convert names to uppercase using `map()`
- Filter students scoring above 60 using `filter()`
- Create a dictionary using dictionary comprehension:
 - Student → Grade
- Create a list of students scoring A grade using list comprehension
- Display the final report